Paper Circuit Signal

Goal: Build a safe paper circuit that uses one LED to communicate a clear signal or message.

35 minutes | 9 steps | Engineering and Design | Playbook B meeting 9, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate the signal three times and identify the exact path electricity follows.

You will use

- Cardstock
- Copper tape
- LEDs
- Switched 2xAA battery holders
- AA batteries
- Clear tape
- Pencils
- Circuit planning cards

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the circuit card on your table.
- Keep the power switch off.

2 Step 2: Power safety — 4 min

- Use only the switched battery holder.
- Keep each battery inside the holder.
- Switch off any warm holder.
- Report the warm holder to the teacher.

3 Step 3: Plan the signal — 5 min

- Choose one clear signal word.
- Draw one complete circuit path.
- Mark which way the long light leg faces.

4 Step 4: Teacher check — 2 min

- Hold up your paper plan.
- Wait until the teacher says yes.
- Do not take a battery box yet.

5 Step 5: Build the circuit — 7 min

- Press copper tape along the path.
- Make each corner smooth.
- Place the LED in the marked direction.

6 Step 6: Power the signal — 5 min

- Connect the holder leads to the marked points.
- Turn on the switch.
- Check the first place where the path breaks.

7 Step 7: Finish the message — 4 min

- Test the light three times.
- Add the signal word after success.

8 Step 8: Power down — 2 min

- Turn off the switch.
- Return the holder by count.

9 Step 9: Exit check

3 min

- Point to the complete circuit path.
- Name one repair you made.

What to turn in

Your signal shown three times and the path traced.

No computer? Another way to do it

Draw the circuit path and trace it with a finger while the teacher demonstrates.

Finished early? Next challenge

Switch Upgrade: Design a paper press switch that turns the signal on only when a user intentionally activates it.

Stay safe

Never connect the battery leads directly. Switch off and report any warm holder immediately. Keep batteries inside the holder and return every holder by count.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.